

Eco-Scan AI

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Project Objectives

- To detect waste items using image recognition.
- To classify waste into categories such as Plastic, Paper, Glass, Metal, Organic and E-Waste.
- To provide proper disposal and recycling guidance.
- To spread awareness about sustainable waste management.
- To use Artificial Intelligence for supporting environmental sustainability.



Problem Statement

Improper waste disposal is one of the major environmental challenges today. Many people are unaware of the correct method of disposing different types of waste. As a result, recyclable materials often end up in landfills, causing pollution and resource wastage.

The aim of this project is to develop an AI-based solution that helps users identify waste items and provides proper recycling and disposal recommendations.

**Plastic Water Bottle Pollution:
Where Do All The Bottles End Up?**



Project Overview - Introduction

Eco-Scan AI is an Artificial Intelligence based waste detection and recycling assistant developed to support sustainable waste management. The system uses Computer Vision and a Convolutional Neural Network (CNN) model to identify different types of waste from uploaded images.

Users can upload an image of a waste item through a simple Streamlit web application. The trained model analyzes the image, predicts the waste category, and provides suitable recycling or disposal guidance. The project aims to encourage responsible waste management practices and create awareness about environmental sustainability through the use of AI technology.

End Users

The proposed system can be used by a wide range of users who want to improve waste segregation and recycling practices.

- Students and educational institutions for learning about waste management.
- Households for proper disposal of daily waste items.
- Recycling centers for quick waste categorization.
- Environmental organizations conducting awareness programs.
- Municipal bodies promoting sustainable waste management.
- General public interested in eco-friendly practices.

The system is designed with a simple interface so that users of all age groups can easily use it.

Wow Factors in Project

Wow Factors

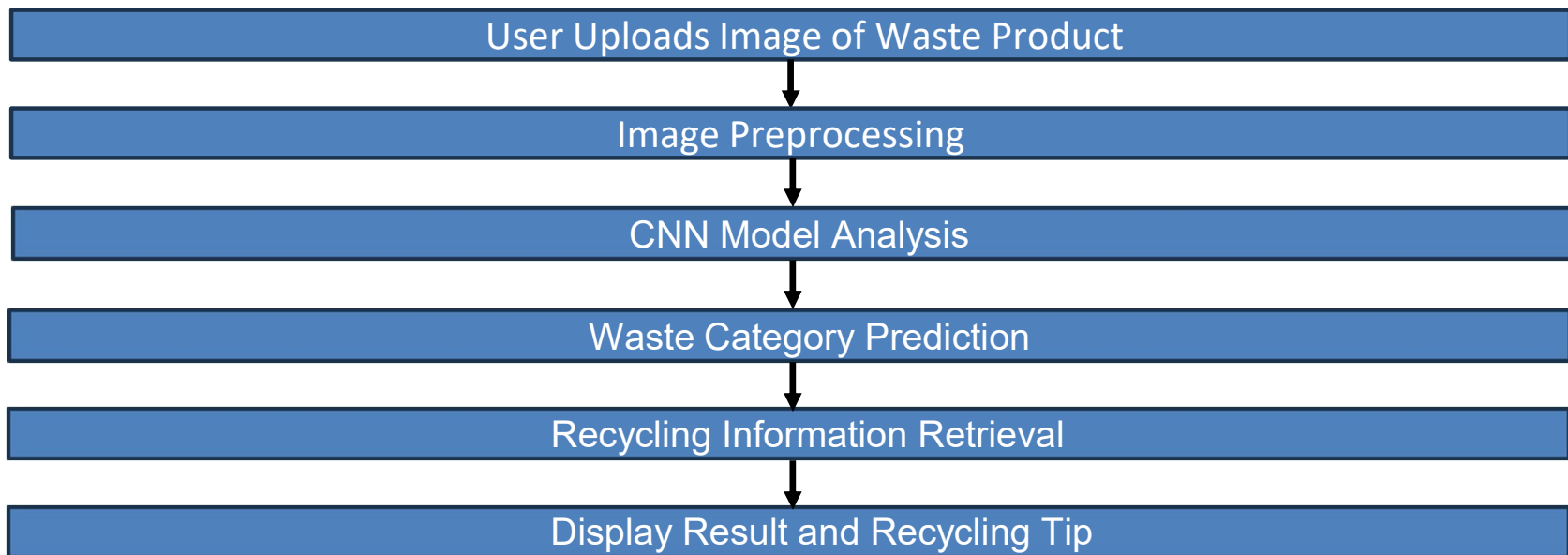
- Uses Artificial Intelligence for automatic waste identification.
- Provides instant waste classification from uploaded images.
- Recommends appropriate recycling or disposal methods.
- Helps users understand correct waste segregation practices.
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- Simple and user-friendly Streamlit-based interface.
- Supports environmental sustainability and green initiatives.
- Can be further enhanced for real-time waste detection applications.

The project combines technology and environmental awareness to solve a real-world problem in a practical way.

Modelling/Block Diagram/Flow of Project

Project Flow

The working process of Eco-Scan AI follows a simple and structured flow.



The uploaded image is first preprocessed and resized according to the model requirements. The trained CNN model then extracts image features and predicts the waste category. Based on the prediction, the system retrieves relevant recycling information and displays the category, disposal method, and recycling tips to the user.

Technology Stack

Programming Language

- Python

Machine Learning Framework

- TensorFlow
- Keras

Web Application Framework

- Streamlit

Supporting Libraries

- NumPy
- Pillow (PIL)

Project Domains

- Artificial Intelligence
- Computer Vision
- Sustainability
- Green Skills

These technologies were selected because they provide an efficient environment for building image classification applications.

Result/Outcomes

Smart Waste Detection & Recycling Assistant

Upload Waste Image

paper227.jpg
9.0KB



Uploaded Image

Detected Waste : Paper

Category

Paper

Disposal Bin

Blue Bin

Recycling Tip

Recycle clean paper.

Confidence

86.48%

Eco-Scan AI

Smart Waste Detection & Recycling Assistant

Upload Waste Image

clothe.jpg
25.9KB



Uploaded Image

Detected Waste : Clothes

Category

Textile

Disposal Bin

Cloth Donation Center

Recycling Tip

Donate usable clothes.

Confidence

98.86%

Result/Outcomes

The Eco-Scan AI application was successfully developed and tested using different waste images. The trained CNN model was able to identify waste categories and provide appropriate disposal and recycling recommendations through the Streamlit web interface.

Key Outcomes

- Successfully classified different types of waste using image recognition.
- Provided suitable disposal bin recommendations.
- Displayed recycling tips for users.
- Developed a simple and user-friendly Streamlit application.
- Promoted awareness about proper waste segregation and recycling.

Sample Results

Paper Waste

- Predicted Category: Paper
- Disposal Bin: Blue Bin
- Confidence: 86.48%

Textile Waste

- Predicted Category: Clothes (Textile)
- Disposal Method: Cloth Donation Center
- Confidence: 98.86%

Conclusion

Eco-Scan AI demonstrates how Artificial Intelligence and Computer Vision can be applied to address environmental challenges. The system helps users identify waste items, understand proper disposal methods, and encourage responsible recycling practices.

The project not only showcases the practical use of machine learning but also supports the goal of creating a cleaner and more sustainable environment. Through this project, we explored the use of AI for solving real-world problems related to waste management and sustainability

Future Perspective

Although the current system successfully classifies waste through uploaded images, there are several opportunities for future enhancement like

- Development of a mobile application for wider accessibility.
- Integration of real-time camera-based waste detection.
- Support for additional waste categories and larger datasets.
- Integration with IoT-enabled smart waste bins.
- Multi-language support for users from different regions.
- Cloud-based deployment for large-scale usage.

These improvements can make the system more efficient, scalable, and useful for practical implementation in smart cities and environmental projects.

Thank You

Github link

https://github.com/PANDYA-RISHI2005/Edunet_Internship_capstone_project

Video (google drive link)

<https://drive.google.com/file/d/1alaFDsMyod0vxBIcMU9A8aqqwpPnFyqi/view?usp=drivesdk>